Toward Personalized Dating Recommendations: Graph-Based Community Detection Using Louvain and Cosine Similarity on OkCupid

1st Aisya Mufidah Najwa
Department of Data Science
Faculty of Informatics, Telkom University
Surabaya, Indonesia
aisyanajwa@student.telkomuniversity.ac.id

2nd Amalia Nur Alifah
Department of Data Science
Faculty of Informatics, Telkom University
Surabaya, Indonesia
*Corresponding Email:
amalialifah@telkomuniversity.ac.id

3rd Tarisma Dwi Putri Alifianti
Department of Data Science
Faculty of Informatics, Telkom University
Surabaya, Indonesia
tarismaalifianti@student.telkomuniversity.ac.id

4th Ikfan Putra Maesru Dwi Pradana
Department of Data Science
Faculty of Informatics, Telkom University
Surabaya, Indonesia
ikfanputramaesrudwip@student.telkomuniversity.ac.id

***Abstract*—The rapid growth of digitalization has transformed social interaction through online dating platforms, which generate complex social networks whose community structure is not directly observable. This paper detects user communities on the OkCupid platform by combining cosine similarity-based network construction with the Louvain algorithm, a pairing chosen for its ability to build a similarity network directly from high-dimensional profile attributes and partition it efficiently at scale. Twelve demographic and lifestyle attributes of 995 California-based users are encoded and standardized, and a weighted network is formed between user pairs whose cosine similarity exceeds 0.7. Executed with a fixed random seed, Louvain partitions the network of 995 nodes and 14,896 edges into ten communities with a modularity of 0.686, in a sparse but locally cohesive graph (density 0.030, clustering coefficient 0.489, average path length 3.72). A sensitivity analysis over thresholds from 0.5 to 0.9 supports 0.7 as the setting balancing modularity, interpretability, and node coverage, and a comparison with four alternative algorithms confirms Louvain is competitive with the best while remaining a widely adopted baseline. Chi-square and ANOVA tests confirm significant differences on every attribute examined, and the profiles reveal segmentation driven jointly by life stage, lifestyle, religious worldview, and identity orientation. The communities provide a reproducible basis for community-aware matching and a potential foundation for recommendation systems, whose full evaluation requires interaction data absent from the dataset.**

Keywords—community detection, dating apps, Louvain algorithm, cosine similarity, social network analysis

I. INTRODUCTION

The growth of digitalization is accelerating rapidly and encompassing various sectors of institutions and daily life. Digitalization has become a connecting force in every aspect of life, particularly in social relationships, reshaping how social ties are formed and removing many of the geographic constraints that once bounded them [1]. Interactions previously limited by physical proximity have moved online, and one visible outcome of this shift is the growth of dating applications. These applications enable users to meet one another, chat, or start dating based on matching preferences and interests of both sides [2], and in practice they generate complex and dynamic social networks whose structures of social relationships are not necessarily visually apparent [3], [4].

One of the major methodologies in social network analysis is community detection, the process of identifying groups within a network whose members are closely connected [5]. Here, a community can represent a group of users with similar preferences or interaction patterns, so community detection provides insight into social structures and aids the development of more personalized features. One of the most efficient and widely used algorithms for this task, particularly in large-scale networks, is the Louvain method, which optimizes modularity, a measure of how well a network is partitioned into communities [6]. Louvain divides the network into communities with high internal and low external connectivity, and its computational efficiency suits the large datasets generated by dating apps [7].

This study applies the Louvain method to detect user communities from dating app profile data, using cosine similarity to build the network, which has already been used for personalized recommendation in other domains such as collaborative filtering [8]. Beyond reporting a single partition, it makes five contributions. The pipeline is reproducible, since a fixed Louvain seed allows the reported partition to be regenerated exactly. The network is characterized through density, clustering coefficient, path length, and community sizes rather than modularity alone. The similarity threshold is justified empirically by a sensitivity analysis. Louvain is compared against alternative algorithms on the same graph. Finally, the communities are interpreted as candidate segments for community-aware recommendation, with an explicit statement of what profile-only data can and cannot validate.

II. LITERATURE REVIEW

A. Previous Research

Research by Thaiprayoon and Unger [9] contributes to our understanding of how user interaction is structured on online platforms through a graph-based approach. They transform a set of user profiles into a graph model to discover meaningful connections and influential users, estimating the semantic

relationship between two profiles with word and sentence embeddings and combining these with personalized PageRank, so that the matchmaking captures complex semantic information and high-order user relationships that simple attribute matching would miss. In a different domain, Sianipar et al. [10] combine sentiment analysis with social network analysis on a public-transport application and find that user discourse is shaped not only by official channels but also by decentralized community interaction, evidence that meaningful community structure can emerge from interaction patterns rather than from demographic attributes alone.

Closer to the dating setting, Felmlee and Kreager [4] analysed the e-mail exchange network of a national dating website and found that aggregated partner preferences give rise to distinct, otherwise invisible clusters of interacting daters, with rated attractiveness and age the strongest correlates of membership, whereas education and race were less common determinants [4]. That study establishes that meaningful community structure exists in online dating networks, but its network was built from observed message exchanges, a form of data that publicly released dating datasets do not contain [4].

Another study by Akhmad Irsyad and Nur Aini Rakhmawati [11] demonstrated community formation on Twitter using cosine similarity and the Louvain method. They constructed a graph based on the similarity of tweet content among users, who were then classified into communities with similar communicative patterns, and showed that Louvain detected communities with high modularity despite the large number of interactions in complex networks [11]. The study, however, focused on a general social media platform rather than dating apps, which feature more intimate social dynamics based on personal preferences.

In parallel, the recommender systems literature has developed reciprocal recommender systems, which account for the two-sided nature of matching: a recommendation succeeds only if both users are interested in each other. RECON, a well-known content-based reciprocal recommender, learns implicit preferences from past contact history and was evaluated on interaction data from a commercial dating site [12]. A recent survey emphasizes that evaluating such systems relies on records of user-to-user interaction from which ground-truth relevance can be derived [13]. Publicly available profile datasets do not satisfy this requirement, which motivates this study's focus on the structural precursor of recommendation: the discovery of latent segments over which recommendations would be made.

Taken together, prior work establishes two results separately: that meaningful community structure exists in dating networks [4], and that cosine similarity with Louvain recovers such structure on general social media [11]. What remains open is the combination this study targets, applying that pipeline to the public profile data of a dating app and reading the resulting communities as a segmentation layer for recommendation. This study addresses that gap on OkCupid data and, rather than claiming a finished recommender, treats the detected communities as a potential foundation for more personalized and adaptive recommendation based on users' social similarities and lifestyles.

B. *Social Network Analysis (SNA)*

Social Network Analysis (SNA) is an analytical approach used to understand the structure and patterns of relationships among entities within a social network. Individuals are represented as nodes and the interactions or relationships between them as edges [14]. A relationship can be represented in two ways: an undirected graph, where each pair of individuals is connected by an edge without an arrow, and a directed graph, where each connection is represented by an arrow that clarifies the direction of the relationship [14]. This study adopts the SNA view directly: each OkCupid user is treated as a node, and an undirected weighted edge is drawn between two users whose profile vectors are sufficiently similar, so that the similarity network analysed in the following sections is itself the graph on which community detection operates.

C. *Louvain Algorithm*

The Louvain method is a community detection algorithm that efficiently optimizes modularity, a measure of how much more densely nodes connect within their assigned communities than would be expected by chance, on large-scale graphs [7].

The algorithm proceeds in two repeated stages. First, each node is moved to the neighbouring community that gives the largest local gain in modularity; once no move improves the score, every community is collapsed into a single node and inter-community weights are summed to form a reduced graph, on which the same step runs again. The two stages alternate until modularity can no longer be increased. Louvain remains widely used, although later work showed it may leave communities that are internally poorly connected and proposed refinements such as Leiden [15]; structurally different alternatives include label propagation [16], greedy modularity maximization [17], and Infomap [18]. Rather than assume Louvain is the best choice, Section IV.D compares these on the same graph.

D. *Cosine Similarity*

Cosine similarity assesses the degree of similarity between two entities represented as feature vectors. Because the twelve profile attributes are categorical, they are first converted to numeric form by label encoding and then standardized, so that each user is described by a numeric vector of comparable scale before any similarity is computed. Similarity is then defined as the cosine of the angle between two such vectors; because it depends on the angle rather than the length, it is insensitive to differences in magnitude, which makes it a standard choice for comparing high-dimensional profile vectors [19].

III. METHODOLOGY

This study follows a structured research pipeline comprising six stages: literature review, data collection, data preprocessing, network construction via cosine similarity, community detection using the Louvain algorithm, and result evaluation. As illustrated in Fig. 1, each stage is designed sequentially to ensure methodological rigor.

A. *Data Collection and Preprocessing*

The dataset used in this study is the publicly available OkCupid profile dataset released by Kim and Escobedo-Land [20]. It consists of the public profiles of 59,946 OkCupid users who lived within 25 miles of San Francisco, active during a period in the 2010s, online within the past year, and with at least one photo. The original authors collected the data from users' public profiles. Messaging history and other non-public information were never accessible to them and are absent from the dataset. No additional web scraping was performed for the

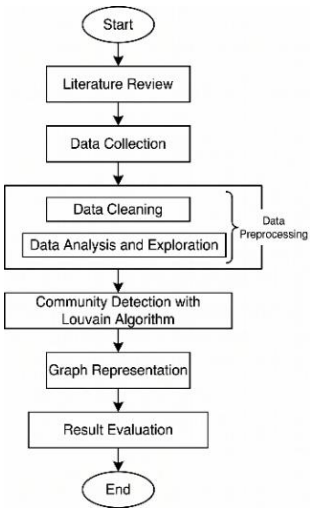


Fig. 1. Research Workflow Diagram.

present study. The analysis relies solely on the published dataset as redistributed by its original authors, and every attribute used here is a profile attribute rather than private communication.

The dataset encompasses structured demographic and lifestyle variables, such as sex, sexual orientation, age (to which the data providers added random noise for anonymization), ethnicity, diet, drinking, and smoking habits, together with semi-structured responses to ten profile essay questions [20]. Only the structured attributes are used here: the essays are excluded because free-text modelling is outside the scope of this work, and ethnicity is excluded to avoid constructing user similarity along an attribute that is both sparsely reported and ethically sensitive in a matching context.

Data cleaning removes rows with missing values rather than imputing them, deletes duplicates, and checks extreme values for input errors. Attribute types are then harmonized, and a correlation matrix over the encoded attributes reveals redundancy between them.

B. Network Construction via Cosine Similarity

Each of the twelve selected attributes is label-encoded and standardized, representing each user as a twelve-dimensional feature vector. Pairwise cosine similarity is computed between all vectors, and the matrix becomes an undirected weighted graph in which an edge is created only when similarity exceeds 0.7, with the similarity value as edge weight. This threshold governs network density and community granularity, and its value is selected empirically in Section IV.C. Encoding, scaling, and similarity computation use scikit-learn [21], and the graph is built with NetworkX [22].

C. Community Detection Using the Louvain Algorithm

The Louvain algorithm is applied to the weighted graph, starting from a partition where every user is its own community, as described in Section II.C [7]. Because the algorithm evaluates nodes in randomized order, different runs can return different partitions of comparable modularity. To ensure exact reproducibility, the algorithm runs with a fixed random seed (random_state = 4082) in the python-louvain implementation, returning an identical partition on every execution. Across ten arbitrary seeds, modularity varies only slightly (mean 0.687, standard deviation 0.003), so partition quality does not depend on the seed chosen.

D. Evaluation Protocol

The evaluation combines four analyses. The partition is characterized by modularity, density, clustering coefficient, average path length on the largest connected component, and community sizes. The similarity threshold is swept from 0.5 to 0.9 to justify the operating point. Louvain is compared on the same graph with Leiden [15], Label Propagation [16], greedy modularity maximization [17], and Infomap[18]. Internal validity is quantified with the silhouette coefficient [23] and the Davies-Bouldin index [24], and the association between membership and each attribute is tested with chi-square tests and, for age, one-way ANOVA. Because the dataset records no user-to-user interaction, no ground truth exists, so external metrics such as NMI or the Adjusted Rand Index and interaction-based recommendation metrics cannot be computed. Section IV.G discusses this constraint.

IV. Results and Discussion

A. Data Preprocessing

The dataset contains 59,946 rows with 31 features. To choose the attributes for similarity, all fields were label-encoded and their pairwise correlation matrix was inspected; from it, twelve attributes were retained that carry social meaning, are reasonably complete, and are not strongly redundant with one another: education, job, religion, orientation, drinks, location, status, diet, offspring, smokes, drugs, and age.

Rows with any missing value among the twelve selected columns were removed rather than imputed, which reduced the data from 59,946 to 9,922 rows. Removal was preferred because imputing categorical profile fields such as religion or offspring would fabricate lifestyle information and could bias the similarity structure the study measures; the trade-off is a smaller sample, so the analysis describes users who completed these fields rather than the entire user base. Age filtering was applied due to extreme values up to 110 years; the IQR-based upper threshold was 53, so only users aged 18 to 53 were kept, and age was grouped into six six-year bands (18-23 through 48-53). The location field, originally in city, state format, was reduced to the state portion. Preprocessing yielded 9,128 rows with twelve columns.

Because pairwise cosine similarity scales as $O(n^2)$ and a readable network visualization needs a tractable node count, a random sample of 1,000 users was drawn from the 9,128 cleaned records with a fixed seed (random_state = 42) so that the draw is reproducible. Category proportions across all twelve attributes differ from the full cleaned dataset by at most 2.85 percentage points, for diet, with most attributes differing by under 1.5 points, confirming the sample is representative before the California restriction. As the study concerns users in California, the sample was then restricted to the 995 California users, which also removes geography as a source of spurious dissimilarity.

B. Network Structure and Community Detection Results

The five users outside California came from Switzerland, New York, Arizona, the Netherlands, and Ohio. In a preliminary run that retained them, they formed two additional outlier communities, raising the count from ten to twelve. Those outliers do not represent meaningful social structure, as their members are very few and separated from the rest of the network almost entirely by location. The analysis therefore continued with 995 users and ten main communities.

This study successfully identified ten communities within the social network formed from twelve attributes and cosine similarity relationships. Only pairs with a similarity score above 0.7 were connected, and community detection was performed using the Louvain algorithm. Fig. 2 visualizes the resulting network, where node colours denote community membership, and Table I summarizes its global structure. The graph is sparse, with a density of 0.030, yet locally cohesive. An average clustering coefficient of 0.489 means that about half of the possible triangles among the neighbours of a user are present, while an average shortest path length of 3.72 within a single component containing all 995 users keeps the network globally reachable. This combination is characteristic of pronounced local groups and is precisely the structure that community detection is designed to expose [5]. The Louvain partition attains a modularity of 0.686, indicating that nodes are substantially more connected within their own communities than would be expected in a random graph with the same degree sequence. The community sizes, given per community in the header of Fig. 3, range from 40 users (4.0%) in C3 to 195 (19.6%) in C0, with a mean of 99.5, so no community dominates, and none is too small to profile.

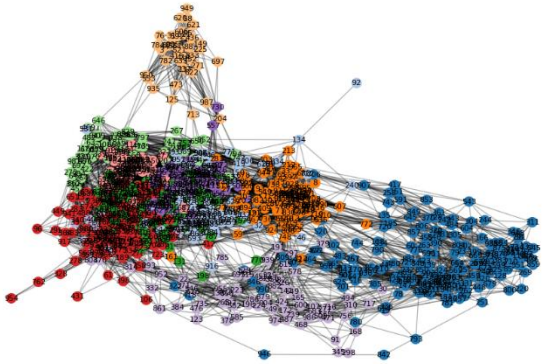


Fig. 2. OkCupid User Community Graph.

TABLE I. GLOBAL NETWORK STATISTICS (THRESHOLD 0.7)

Metric	Value	Metric	Value
Nodes	995	Path length (LCC)	3.72
Edges	14,896	Size of LCC	995
Network density	0.030	Communities	10
Clustering coeff.	0.489	Modularity	0.686

C. *Threshold Sensitivity Analysis*

The similarity threshold is the only free parameter of network construction, so a natural question is whether the reported structure is an artefact of the value chosen. Table II sweeps it from 0.5 to 0.9 on the same similarity matrix, revealing two opposite failure modes. At 0.5–0.6 the graph becomes dense (44,740 and 27,445 edges). Community boundaries blur as almost every pair connects, and modularity falls to 0.528–0.613. At 0.8–0.9 modularity rises to 0.774–0.887, but the partition fragments into 49 and 339 communities. This isolates 32 and 258 users with no neighbours at all. High modularity here comes from fragmentation, not meaningful structure. Two criteria guided the choice of 0.7. The first is coverage. At 0.7, every user still remains connected, the largest value at which that holds. The second is interpretability. Its ten communities of 40–195 members can each be profiled, unlike the 49 groups at 0.8 that average only twenty members. A deployed system could tune the threshold differently. For the descriptive goals here, 0.7 satisfies both criteria at once.

TABLE II. SENSITIVITY TO THE COSINE SIMILARITY THRESHOLD

Thr.	Edges	Density	Comm.	Mod.	Isolated
0.5	44,740	0.091	9	0.528	0
0.6	27,445	0.056	9	0.613	0
0.7	14,896	0.030	10	0.686	0
0.8	6,452	0.013	49	0.774	32
0.9	1,585	0.003	339	0.887	258

D. *Comparison with Alternative Algorithms*

To verify that Louvain is an appropriate choice rather than merely a popular one, it was compared with four alternative algorithms on the same graph. Leiden is a refinement of Louvain that guarantees well-connected communities [16]. Label Propagation spreads labels without optimizing an explicit objective [17]. Greedy modularity maximization (CNM) is an agglomerative method [18]. Infomap is a flow-based method that optimizes a description-length objective rather than modularity [19]. All were executed with the same random seed for reproducibility. As Table III shows, Leiden and Louvain give almost identical partitions. Both recover ten communities at the highest modularity, 0.689 and 0.686, with Leiden marginally ahead. Label Propagation is fast but fragments the network into sixteen communities at a lower modularity (0.602). CNM merges the users into only six coarse communities (0.593) at the slowest runtime. Infomap collapses the network into three communities with a much lower modularity (0.267), reflecting that its flow-based objective does not target modularity. Louvain therefore sits with Leiden at the top of the ranking while remaining the most widely adopted baseline, which supports its use here.

TABLE III. COMPARISON OF COMMUNITY DETECTION ALGORITHMS

Algorithm	Communities	Modularity	Time (s)
Leiden	10	0.689	0.024
Louvain	10	0.686	0.254
Label Propagation	16	0.602	0.044
Greedy modularity	6	0.593	1.535
Infomap	3	0.267	0.066

The near-tie between Louvain and Leiden is consistent with Leiden being a direct refinement of Louvain. Leiden attains a slightly higher modularity while guaranteeing well-connected communities, so it is a promising direction for future work. The present study is deliberately scoped to Louvain as the established baseline, and a fuller Leiden-based analysis is left to subsequent work.

E. *Validity of the Detected Communities*

In the feature space, the partition attains a silhouette coefficient of 0.248 and a Davies-Bouldin index of 1.622, indicating moderate rather than strong geometric separation. This is expected: Louvain optimizes modularity on a thresholded similarity graph, whereas both indices measure compactness in the feature space, so a partition can be strongly modular while remaining geometrically overlapping. Chi-square tests between community membership and each categorical attribute are significant at $\alpha = 0.05$ for all ten attributes (χ^2 ranging from 407.90 to 918.77, all $p < 10^{-37}$). Age also differs across communities (one-way ANOVA, $F = 69.74$, $p = 3.06 \times 10^{-99}$). Because the communities are derived from these same attributes, the tests are diagnostic rather than independent confirmation. They establish that no attribute is

irrelevant and that membership is associated with the full attribute set rather than driven by a single dominant variable.

F. Discussion

As illustrated in Fig. 3, the ten detected communities exhibit substantial variation across six key attributes: average age, secularity, drug use, smoking, heavy drinking, and sexual orientation. Rather than describing each community in isolation, this section examines how they differ along these features and what that implies for personalized recommendation. In the figure, cell colour is normalized within a row, so intensity expresses how a community compares with the others on that attribute, while the printed value reports the underlying percentage or mean.

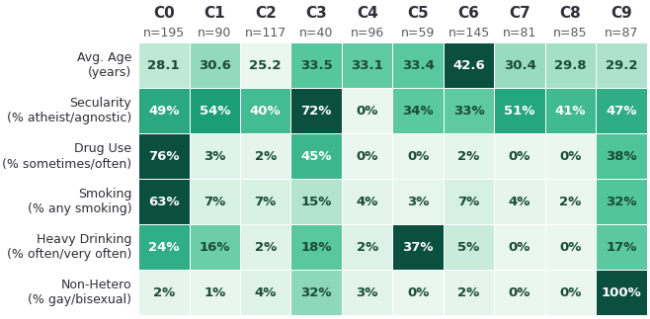


Fig. 3. Community Characteristics Heatmap Across 10 Communities.

For ease of reference, each community is additionally given a representative label reflecting its dominant demographic and behavioural profile: C0 (Young Liberal), C1 (Young Professionals), C2 (Students), C3 (Creative Adults), C4 (Urban Professionals), C5 (Conservative), C6 (Mature Families), C7 (Tech Workers), C8 (Religious Professionals), and C9 (LGBTQ+ Youth). A clear life-stage segmentation emerges along the age dimension. Communities C2 (25.2 years), C0 (28.1), C9 (29.2), and C8 (29.8) occupy the youngest band and are populated predominantly by students and early-career individuals still exploring their identity: 65 of the 117 members of C2 report being students and 52 report working on a college degree, and C9 shows a similar composition. Communities C7 (30.4), C1 (30.6), C4 (33.1), C5 (33.4), and C3 (33.5) occupy a middle band associated with career and lifestyle stability. Community C6 stands out as a clear outlier at 42.6 years and is the only community in which parenthood is the norm, its two most frequent offspring categories being having a kid (30 members) and having kids (27 of 145). This implies that matching algorithms should weight age-group compatibility before applying behavioural similarity scores.

The lifestyle spectrum, captured across drug use, smoking, and heavy drinking, reveals two distinct poles. Community C0 consistently occupies the liberal end across all three dimensions (76% drug use, 63% reporting some form of smoking, 24% heavy drinking), reflecting an open, experimental urban lifestyle characteristic of younger, secular demographics. At the opposite pole, communities C4, C7, and C8 record 0% drug use with near-zero smoking and heavy drinking, forming a cluster of health-conscious professionals, with C7 additionally concentrated in technology-related occupations. Community C5 presents an interesting paradox, despite no drug use and only 3% smoking, it records the highest heavy drinking rate (37%). Its drinking distribution is in fact strongly bimodal, with 35 of its 59 members reporting that they do not drink at all and 22 that they drink often, so alcohol use in this group is polarized rather than uniformly permissive. Such within-spectrum nuances are the kind of behavioural signal that cosine similarity captures effectively.

Religious orientation shows a polarized pattern that correlates with age and lifestyle. The younger, more lifestyle-liberal communities, particularly C3 (72%), C1 (54%), C7 (51%), C0 (49%), and C9 (47%), exhibit high rates of atheism and agnosticism, consistent with secularization tendencies among urban millennials in the San Francisco area, notably C3, whose members are largely employed in creative fields such as arts and writing. Communities C8 and C6, which are older or more conservative, show higher affiliations with Catholicism and Christianity: these are the two most frequent religions in C8 (25 and 21 of 85 members) and rank immediately behind agnosticism in C6. Community C4 is the most distinctive case, containing no atheist or agnostic members at all and instead dominated by the other category (65 of 96) together with Judaism (29), a concentration of religious identities outside the Christian and secular mainstream.

The most structurally distinct community is C9, the only group composed entirely of non-heterosexual individuals (100% gay or bisexual: 54 bisexual and 33 gay members). C9 overlaps with C0 on several dimensions, including a comparable average age (29.2 versus 28.1 years), a similar secular outlook (47% versus 49%), and non-negligible substance use (38% drug use, 32% smoking), yet forms a separate community. This shows that cosine similarity over a multidimensional attribute vector captures identity-related differences that demographics-based filters alone cannot detect, and implies that C9 users are best served by intra-community matching rather than cross-community suggestions toward C0, despite their demographic proximity.

Taken together, the profiles reveal that social segmentation within online dating platforms is multi-causal. Affiliation cannot be attributed to any single characteristic. Instead it reflects combinations of age stage, lifestyle conservatism, religious worldview, and identity orientation, consistent with the significance tests of Section IV.E. This supports high-dimensional similarity metrics over single-attribute filters for community-aware recommendation, since each of the ten communities constitutes a distinct segment with internally compatible preferences and a principled basis for targeted candidate generation. This is a structural argument rather than a measured improvement in recommendation quality, addressed next.

G. Limitations and Future Work

Three limitations qualify these results. First, the analysis uses 995 users from a single platform, metropolitan area, and period, so the specific communities should not be generalized, though the pipeline is transferable. Second, the twelve categorical attributes were label-encoded before standardization, imposing an arbitrary ordinal scale on nominal categories. The bimodal drinking profile of C5 is a visible consequence, and one-hot encoding, Gower distance, or embedding-based representations would remove this artefact. Third, the dataset contains only static profile attributes, with no record of likes, matches, or messages. Precision@K, Recall@K, and NDCG require ground-truth relevance from observed interaction, and reciprocal recommendation requires evidence of interest from both sides of a pair [12], [13]. Neither is recoverable from profile data, so proxy labels invented from the profiles would measure the

assumption rather than test it. The contribution here is the reproducible identification of latent user segments, the candidate generation layer a reciprocal recommender consumes. Future work will apply the pipeline to interaction-rich data to evaluate candidate generation against reciprocal baselines such as RECON [12].

V. Conclusion

This study used the Louvain method to discover user communities on OkCupid, based on cosine similarity over demographic and behavioural characteristics. From a sample of 995 users, ten communities emerged within a network of 14,896 weighted edges, showing high internal connectivity alongside heterogeneity in demographics, lifestyle, and preferences, a segmentation confirmed statistically for every attribute examined.

Community cohesion is reflected in a modularity of 0.686 within a sparse (density 0.030) yet locally cohesive network (clustering coefficient 0.489, path length 3.72). The 0.7 threshold is justified by a sweep from 0.5 to 0.9, where lower values blur community boundaries and higher values fragment the network and isolate users. Louvain's choice is supported by a comparison against four alternatives, tying with Leiden for the highest modularity and clearly outperforming Label Propagation, greedy modularity maximization, and Infomap, with all results reproducible under a fixed seed. Since community members share similar preferences and lifestyles, this mapping can serve as a basis for recommendation systems favouring social-characteristic similarity. The study stops at community detection: with no interaction records available, the recommendation layer remains an implication of community structure, its empirical evaluation left to future work on interaction-rich data.

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